WEEK 1: DATA ANALYSIS AND PYTHON QUICK PRACTICE SHEET

A short preparation resource — not a mock test

Topics to Revise

- Reading business metrics such as totals, averages, percentages and rates.
- Using data to explain a business change and asking what other information may be needed.
- Basic relationships between tables and why a single order can contain multiple items.
- Python basics: lists, loops, if conditions, functions.
- Basic Pandas operations: selecting rows/columns and filtering data.
- Statistics concepts: mean, outliers, random samples and the Central Limit Theorem.
- Understanding the purpose of a hypothesis test and interpreting a p-value in plain language.

How to use this sheet: Try these few questions to revise the broad skills used in data analysis. This sheet intentionally does not match the assessment's length, question count, structure, or exact question styles.

Practice Questions

1. Making sense of a business metric

A café served 200 customers on Monday and 260 on Tuesday. Its total sales were ₹30,000 on Monday and ₹33,800 on Tuesday.

a) Calculate sales per customer for each day.

b) Sales increased on Tuesday. Give one possible reason why this does not necessarily mean the café performed better in every respect.

c) Suggest one additional metric you would check.

2. Working with simple tables

A shop has an Orders table with one row per order and an Items table with one row per item in an order.

a) If order O15 contains three items, how many rows might O15 have in the Items table?

b) Why could counting rows in the Items table give a larger number than counting orders?

c) Name a field that could be used to connect the two tables.

3. Python reasoning

Consider the code:

numbers = [2, 4, 6]\nfor n in numbers:\n   if n > 3:\n   print(n)

a) What will be printed?

b) Change the condition so that only numbers greater than 4 are printed.

c) In your own words, what does a for loop do?

4. A small data-cleaning task

A list contains these values: ["12", "8", "missing", "5"].

a) Which values can be converted directly to integers?

b) Why might a data-cleaning program need to handle invalid values instead of stopping completely?

c) Write a short Python example using try/except to handle one invalid conversion.

5. Basic Pandas practice

Assume a DataFrame named sales has columns customer, city and amount.

a) Write code to display only the amount column.

b) Write code to select rows where amount is greater than 1,000.

c) In simple terms, how is selecting a row by its label different from selecting a row by its position?

6. Thinking statistically

A company records the delivery time of every order. Most deliveries take around 30 minutes, but a few take more than 90 minutes.

a) What does this tell you about the distribution of delivery times?

b) Should the company automatically delete the very long delivery times? Explain.

c) If the company takes several random samples and calculates the average delivery time for each sample, what is being studied: individual delivery times or a distribution of sample averages?

7. Interpreting a test result

A company investigates whether its late-delivery rate is higher than its target. A statistical test reports a p-value of 0.03. The chosen significance level is 0.05.

a) Based on this threshold, what decision would typically be made about the null hypothesis?

b) Does this result explain why deliveries are late or identify the best solution? Why or why not?

Suggested Revision Approach

- Explain your reasoning in plain language before doing calculations.
- Practise writing short Python and Pandas snippets by hand.
- For each business metric, be clear about what is counted in the numerator and denominator.
- For statistics questions, focus on interpretation and limitations rather than memorising formulas alone.